

End-to-End Pharmaceutical Analytics & Forecasting Framework – MECE Breakdown

Sales & Revenue Analysis

Category	Description
Revenue Trend Analysis	Analyze monthly and yearly revenue trends to identify growth patterns and sales fluctuations.
Medicine Category Analysis	Evaluate which pharmaceutical categories contribute most to revenue generation.
Country-wise Sales Analysis	Compare pharmaceutical sales performance across different countries and regions.
Warehouse Revenue Analysis	Identify warehouses contributing highest sales and operational performance.

Inventory & Stock Monitoring Analysis

Category	Description
Stock Level Analysis	Monitor medicine inventory levels across warehouses to identify stock distribution patterns.
Reorder Requirement Analysis	Detect medicines below reorder threshold requiring replenishment.
Inventory Heatmap Analysis	Analyze warehouse-wise inventory concentration using heatmap visualization.
Stock Optimization Analysis	Evaluate inventory balancing opportunities across warehouse locations.

Expiry & Risk Analysis

Category	Description
Expiry Risk Monitoring	Identify medicines approaching expiry and categorize operational risk levels.
Days-to-Expiry Analysis	Analyze remaining shelf life of medicines to improve inventory planning.
High-Risk Inventory Analysis	Detect inventory with immediate expiry risk requiring operational action.
Inventory Safety Monitoring	Evaluate medicine safety and stock sustainability through expiry tracking.

Supplier & Supply Chain Analysis

Category	Description
Supplier Delay Analysis	Analyze supplier delivery delays to identify operational inefficiencies.
Lead Time Analysis	Evaluate supplier lead time performance and delivery consistency.
Supplier Risk Assessment	Categorize suppliers into risk levels based on operational performance metrics.
Supply Chain Efficiency Analysis	Examine supplier operations impacting pharmaceutical inventory continuity.

Statistical & Time-Series Analysis

Category	Description
Trend Analysis	Analyze long-term pharmaceutical sales patterns using time-series techniques.
Seasonality Analysis	Identify recurring seasonal demand fluctuations in medicine sales.
Correlation Analysis	Measure relationships between sales volume, revenue, and operational metrics.
Moving Average Analysis	Smooth short-term fluctuations to identify long-term revenue trends.

Forecasting & Predictive Analysis

Category	Description
ARIMA Forecasting	Predict future pharmaceutical sales using ARIMA forecasting models.
Sales Forecast Visualization	Compare actual and forecasted sales performance trends.
Demand Prediction Analysis	Estimate future medicine demand patterns using statistical forecasting.
Forecast Trend Evaluation	Analyze expected future growth and operational demand patterns.